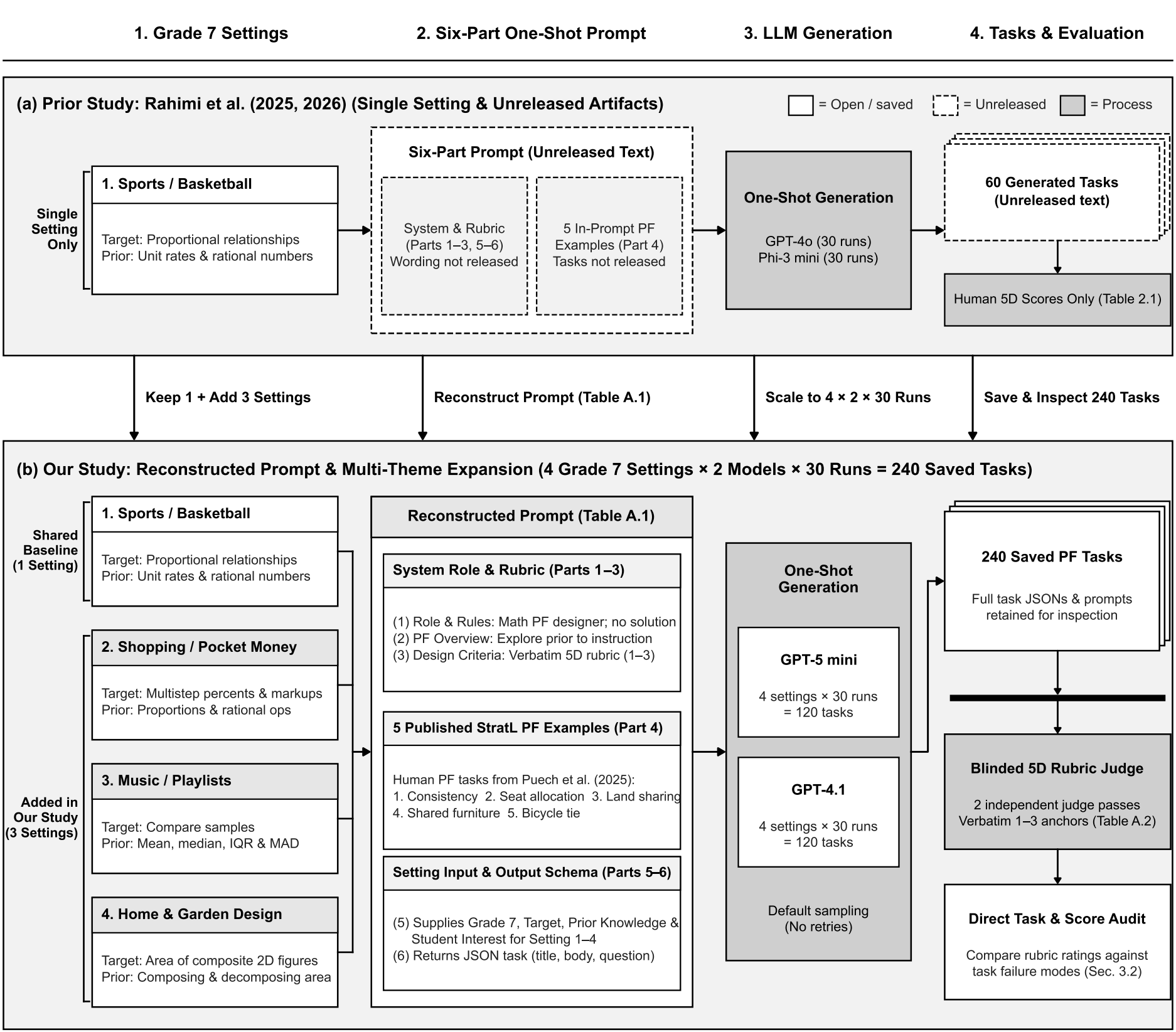
1. Grade 7 Settings
2. Six-Part One-Shot Prompt
3. LLM Generation
4. Tasks & Evaluation
(a) Prior Study: Rahimi et al. (2025, 2026) (Single Setting & Unreleased Artifacts)
= Open / saved
= Unreleased
= Process
Single Setting Only
1. Sports / Basketball
Target: Proportional relationships
Prior: Unit rates & rational numbers
Six-Part Prompt (Unreleased Text)
System & Rubric (Parts 1–3, 5–6) Wording not released
5 In-Prompt PF Examples (Part 4) Tasks not released
One-Shot Generation
GPT-4o (30 runs)
Phi-3 mini (30 runs)
60 Generated Tasks (Unreleased text)
Human 5D Scores Only (Table 2.1)
Keep 1 + Add 3 Settings
Reconstruct Prompt (Table A.1)
Scale to 4 × 2 × 30 Runs
Save & Inspect 240 Tasks
(b) Our Study: Reconstructed Prompt & Multi-Theme Expansion (4 Grade 7 Settings × 2 Models × 30 Runs = 240 Saved Tasks)
Shared Baseline (1 Setting)
1. Sports / Basketball
Target: Proportional relationships
Prior: Unit rates & rational numbers
Added in Our Study (3 Settings)
2. Shopping / Pocket Money
Target: Multistep percents & markups
Prior: Proportions & rational ops
3. Music / Playlists
Target: Compare samples
Prior: Mean, median, IQR & MAD
4. Home & Garden Design
Target: Area of composite 2D figures
Prior: Composing & decomposing area
Reconstructed Prompt (Table A.1)
System Role & Rubric (Parts 1–3)
(1) Role & Rules: Math PF designer; no solution
(2) PF Overview: Explore prior to instruction
(3) Design Criteria: Verbatim 5D rubric (1–3)
5 Published StratL PF Examples (Part 4)
Human PF tasks from Puech et al. (2025):
1. Consistency 2. Seat allocation 3. Land sharing
4. Shared furniture 5. Bicycle tie
Setting Input & Output Schema (Parts 5–6)
(5) Supplies Grade 7, Target, Prior Knowledge & Student Interest for Setting 1–4
(6) Returns JSON task (title, body, question)
One-Shot Generation
GPT-5 mini
4 settings × 30 runs = 120 tasks
GPT-4.1
4 settings × 30 runs = 120 tasks
Default sampling (No retries)
240 Saved PF Tasks
Full task JSONs & prompts retained for inspection
Blinded 5D Rubric Judge
2 independent judge passes
Verbatim 1–3 anchors (Table A.2)
Direct Task & Score Audit
Compare rubric ratings against task failure modes (Sec. 3.2)